

Microfoundations I: Consumption and Saving

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Keynes (1936) on Consumption

The fundamental psychological law, upon which we are entitled to depend with great confidence both a priori from our knowledge of human nature and from the detailed facts of experience, is that men are disposed, as a rule and on the average, to increase their consumption as their income increases but not by as much as the increase in the income.



Keynes (1936) on Consumption

This means that: $C = c(Y)$, with $0 < c'(Y) < 1$ (marginal propensity to consume)

- $$\frac{c'(Y)Y}{c(Y)} = \frac{\partial \log(c(Y))}{\partial \log(Y)} \in (0, 1)$$

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Keynes (1936) on Consumption

- A set of navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.



Keynes (1936) on Consumption

- Given that consumption has been close to a constant of the US GDP (65%)
- $C = 0.65Y$ implies $\frac{c'(Y)Y}{c(Y)} = 1$ (as we see in the US data)
- This evidence is that countries do *not* consume a lower fraction of their income as they grow richer
- Why do cross-section and time series data look so different?



Intertemporal Consumption Decision: 2-period Model

- Household lives for 2 periods
- In period 1 it receives income y_1 and in period 2 y_2
- Household has to decide how much to consume in period 1, taking into account that saving means future consumption
- Household's preferences:

$$U(c_1, c_2) = u(c_1) + \beta u(c_2)$$

with $0 < \beta < 1$ being the time discount factor (impatience)

- Household's (period-by-period) budget constraints:

$$a = y_1 - c_1 \quad (1)$$

$$c_2 = y_2 + (1 + r)a \quad (2)$$



Intertemporal Consumption Decision: 2-period Model

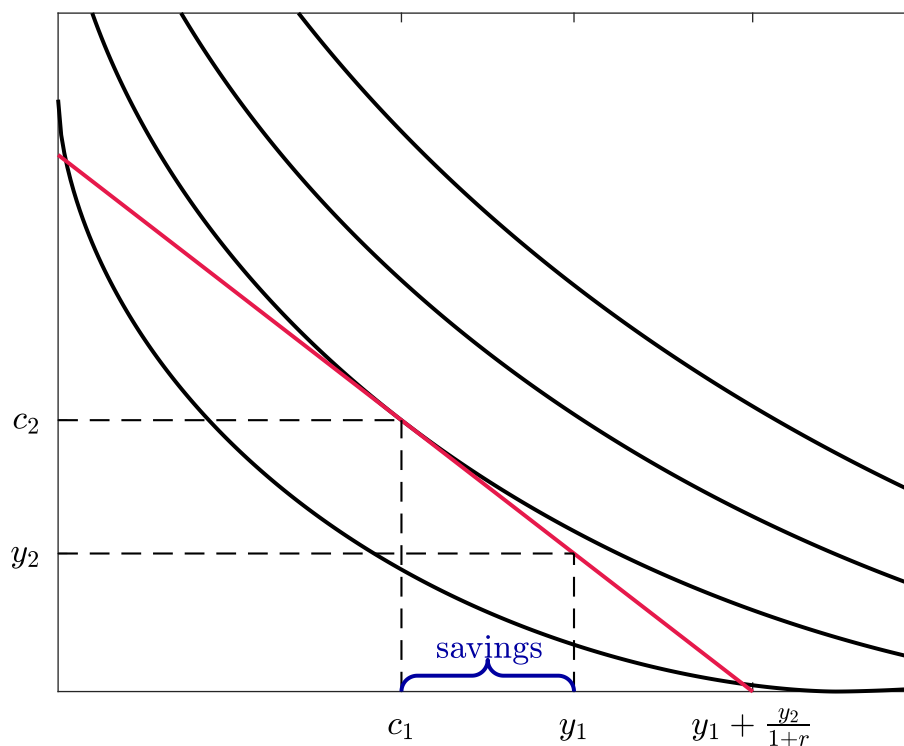
- Can a be also negative?
- Only if there are no borrowing constraints (and no default), which is what we assume for now
- We can combine the budget constraints (1) and (2) into the intertemporal budget constraint:

$$c_1 + \frac{1}{1+r} \cdot c_2 = y_1 + \frac{1}{1+r} \cdot y_2 \quad (3)$$

- This budget constraint has slope $-(1+r)$ (higher r means steeper BC) and goes through the point (y_1, y_2)
- Households maximize the utility function subject to (3)

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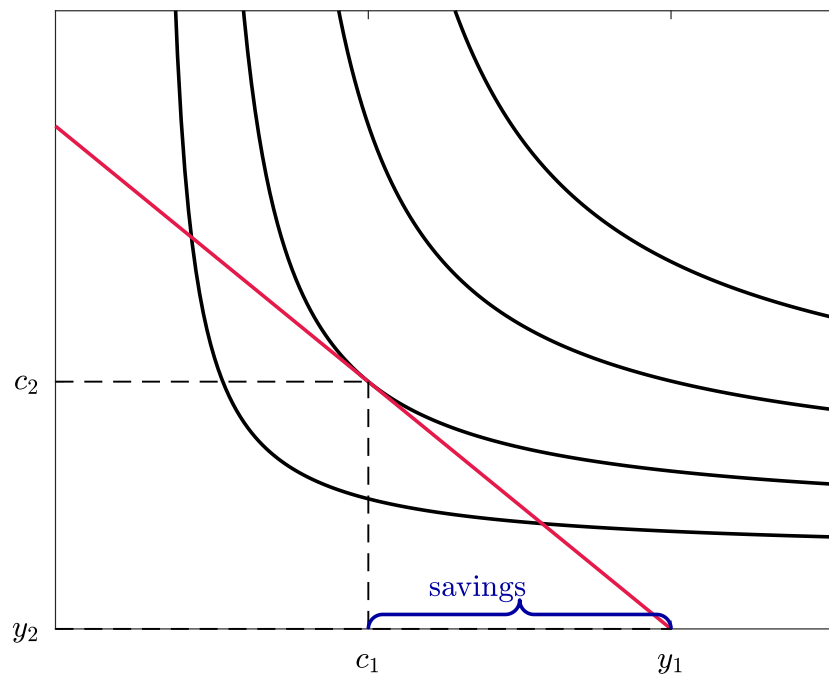
Intertemporal Consumption Decision: 2-period Model



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Intertemporal Consumption Decision: 2-period Model

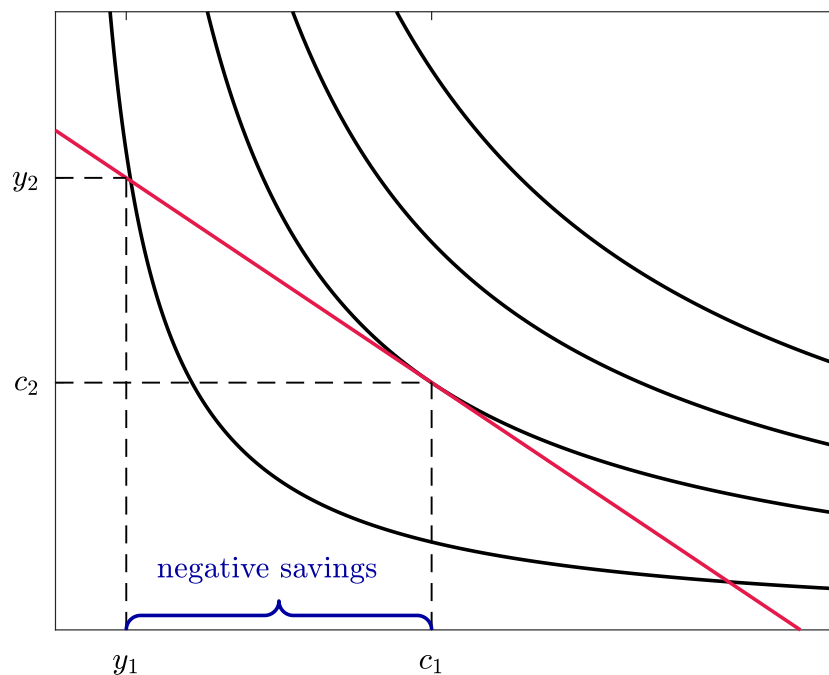
- Retirement example: $y_1 > 0$ and $y_2 = 0$, so agents will choose $a = y_1 - c_1 > 0$



Navigation icons: back, forward, search, etc.

Intertemporal Consumption Decision: 2-period Model

- Higher future income example: low y_1 and expectations of high y_2 makes individuals choose $a = y_1 - c_1 < 0$



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Intertemporal Consumption Decision: 2-period Model

- First Order Condition (“Euler Equation”):

$$u'(c_1) = \beta(1 + r)u'(c_2) \quad (4)$$

- This corresponds to the tangency point of the previous figures:

$$\frac{u'(c_1)}{\beta u'(c_2)} = (1 + r)$$



Intertemporal Consumption Decision: 2-period Model

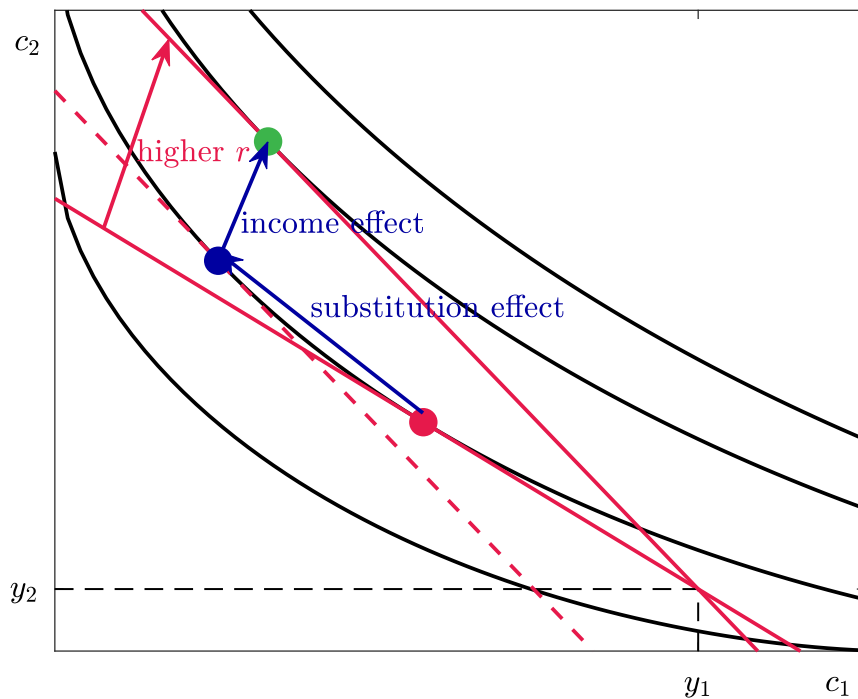
- **Effects of Interest Rate Increase:**

- ▶ Budget constraint continues to cross the point (y_1, y_2) , but it is steeper
- ▶ There is a **substitution effect**: lower c_1 and increase a
- ▶ There is an **income effect**: depends on agent being a lender or a borrower



Intertemporal Consumption Decision: 2-period Model

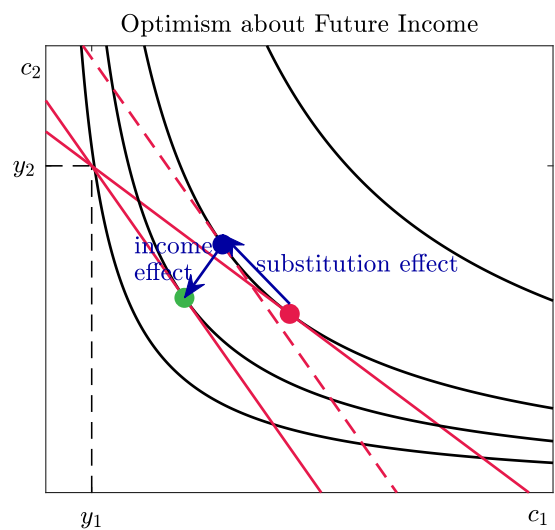
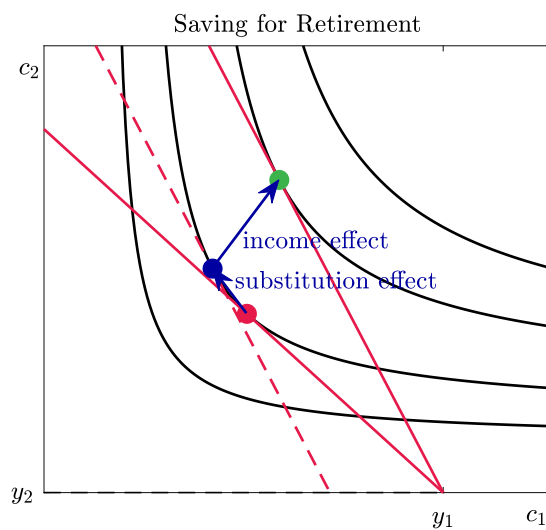
- **Effects of Interest Rate Increase:** $a > 0$



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Intertemporal Consumption Decision: 2-period Model

- **Effects of Interest Rate Increase:**



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Intertemporal Consumption Decision: 2-period Model

- **The Permanent Income Hypothesis:** Friedman (1957)
- Idea captured in the 2-period model
- Consumption depends on total value of household's income over time, instead of current income
- How much does consumption increase after an increase in income? It depends!
 - ▶ Temporary increase: consumption increases but less than the increase in income (as in Keynes)
 - ▶ Permanent increase: much higher increase in consumption
- PIH reconciles cross-section and time-series evidence on consumption:
 - ▶ household with *temporarily* low income will have high c/y (permanent > temporary income)
 - ▶ household with *temporarily* high income will have low c/y (permanent > temporary income)



Intertemporal Consumption Decision: 2-period Model

- PIH reconciles cross-section and time-series evidence on consumption:
 - ▶ **cross-section:** high (low) income data contains high (low) permanent income and high (low) temporary income (with not so high (low) permanent income); on average, households with higher total income are more likely to have had a better-than-usual year, thus lower c/y .
 - ▶ **time series:** adding up many people over longer periods, temporary components average out; average total income is closer to permanent income, so elasticity of consumption is closer to 1.



Intertemporal Consumption Decision: 2-period Model

- **Taxes and Ricardian Equivalence:** how does consumption react to taxes?
 - ▶ Suppose government spending of G_1 and G_2 , exogenously given
 - ▶ Government collects lump sum taxes τ_1 and τ_2
 - ▶ Government can borrow and lend at interest rate r
 - ▶ Government instantaneous budget constraints:

$$B = G_1 - \tau_1 \quad (5)$$

$$\tau_2 = G_2 + (1 + r)B \quad (6)$$

- Using (5) to replace B in (6), we obtain the government intertemporal budget constraint:

$$G_1 + \frac{1}{1+r} \cdot G_2 = \tau_1 + \frac{1}{1+r} \cdot \tau_2 \quad (7)$$



Intertemporal Consumption Decision: 2-period Model

- **Taxes and Ricardian Equivalence:** how does consumption react to taxes?
- We need to adapt household's BC because now they can consume out of income net of taxes:

$$c_1 + \frac{1}{1+r} \cdot c_2 = y_1 - \tau_1 + \frac{1}{1+r} \cdot (y_2 - \tau_2) \quad (8)$$

- Given that (7) specifies that the present value of taxes equal the present value of government spending, we can use this to substitute the present value of taxes in the household's BC, to obtain:

$$c_1 + \frac{1}{1+r} \cdot c_2 = y_1 + \frac{1}{1+r} \cdot y_2 - \left(G_1 + \frac{1}{1+r} \cdot G_2 \right) \quad (9)$$

- Any combination of (τ_1, τ_2) that satisfy (7) is equivalent for the household, i.e., the timing of taxes doesn't matter [**Ricardian Equivalence**]



Intertemporal Consumption Decision: 2-period Model

- **Taxes and Ricardian Equivalence:** how does consumption react to taxes?
- In order for the Ricardian Equivalence to hold, we need:
 - ▶ perfectly rational agents that understand (7)
 - ▶ change of taxes doesn't change expectations of future government spending
 - ▶ common interest rate, r , for households and government
 - ▶ lump sum taxes



Intertemporal Consumption Decision: 2-period Model

- **Precautionary Savings:** what happens when we introduce *uncertainty* in the model?
- Let's consider two households:
 - ▶ *Household A*: no uncertainty, (y_1, y_2) known
 - ▶ *Household B*: y_1 ; uncertainty about y_2 : $y_2 + \epsilon$ or $y_2 - \epsilon$, probability 50%
 - ▶ On average, both households make the same lifetime income
- Choices of *Household A* are the same as before
- *Household B* in period 1: $a = y_1 - c_1$. This savings can lead to two levels of c_2

$$c_2^H = y_2 + \epsilon + (1 + r)(y_1 - c_1)$$

$$c_2^L = y_2 - \epsilon + (1 + r)(y_1 - c_1)$$



Intertemporal Consumption Decision: 2-period Model

- **Precautionary Savings:** what happens when we introduce *uncertainty* in the model?

- Household B solves:

$$\text{Max}_{c_1} u(c_1) + \beta \left[\frac{1}{2} u(y_2 + \epsilon + (1+r)(y_1 - c_1)) + \frac{1}{2} u(y_2 - \epsilon + (1+r)(y_1 - c_1)) \right]$$

- FOC:

$$\begin{aligned} u'(c_1) - \beta(1+r) \left[\frac{1}{2} u'(y_2 + \epsilon + (1+r)(y_1 - c_1)) + \right. \\ \left. \frac{1}{2} u'(y_2 - \epsilon + (1+r)(y_1 - c_1)) \right] = 0 \\ \Rightarrow u'(c_1) = \beta(1+r) \left[\frac{1}{2} u'(c_2^H) + \frac{1}{2} u'(c_2^L) \right] \end{aligned} \quad (10)$$

- Will uncertainty make B save more or less?



Intertemporal Consumption Decision: 2-period Model

Proposition (1)

If $u'(c)$ is a strictly convex function, then B saves **more** than A .

Proof.

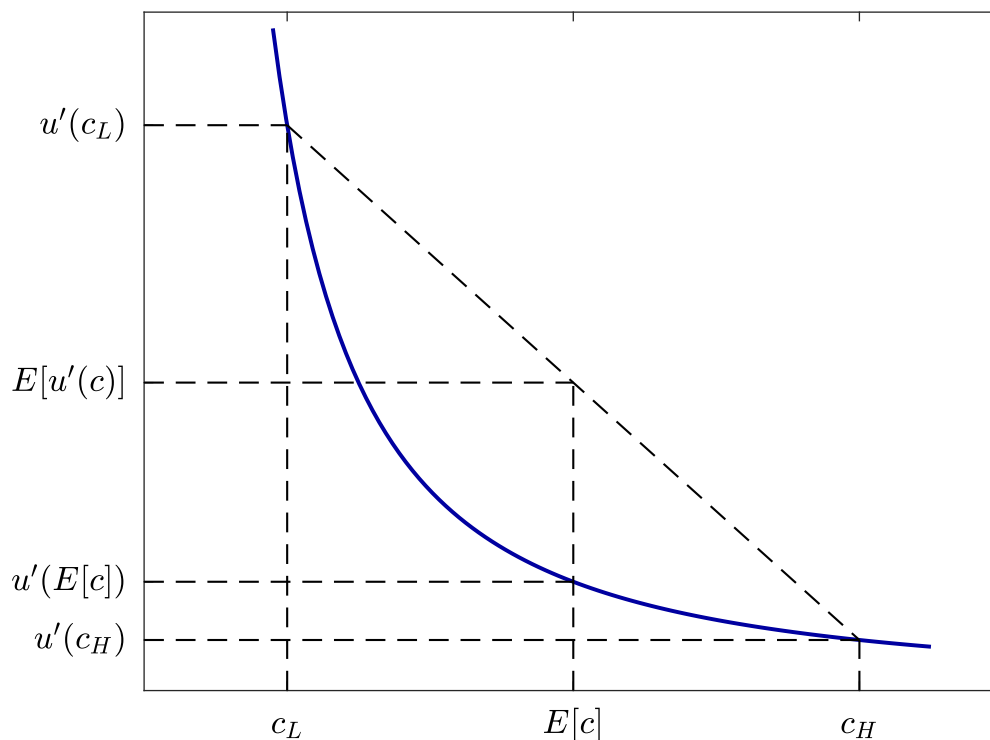
Assume the contrary, i.e. $c_1^B \geq c_1^A$. Then:

$$\begin{aligned} u'(c_1^B) &\leq u'(c_1^A) \\ \beta(1+r) \left[\frac{1}{2} u'(c_2^H) + \frac{1}{2} u'(c_2^L) \right] &\leq \beta(1+r) u'(c_2^A) \\ \left[\frac{1}{2} u'(c_2^H) + \frac{1}{2} u'(c_2^L) \right] &\leq u'(c_2^A) \\ \frac{1}{2} c_2^H + \frac{1}{2} c_2^L &> c_2^A \\ y_2 + (1+r)(y_1 - c_1^B) &> y_2 + (1+r)(y_1 - c_1^A) \\ c_1^B &> c_1^A \end{aligned}$$

which is a contradiction.



Intertemporal Consumption Decision: 2-period Model



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Intertemporal Consumption Decision: Behavioral Theories

- We presented a theory of **rational** consumption-savings decisions
- Non-rational behavior:
 - ▶ 401(k) plans: tax-advantaged plans offered by firms to employees; employees contribute a fraction of their salary to be withdrawn upon retirement, with returns tax free. Choi et al. (2004): changing default options changed employees' decisions.
 - ▶ Rational inattention: paying attention is costly, so households rationally choose what they will pay attention and what they will ignore.
 - ▶ Poor self-control: buy a good they like whenever they encounter it, so people don't think about the future.
 - ▶ Hyperbolic Discounting ("Present bias"): decreasing discount rates as time elapses; leads to decisions that are dynamically inconsistent.

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